



Design Challenge Report

Self-Biased Sub-1V Bandgap Reference

Analog IC Design | Cadence Virtuoso 65nm

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Technology	65 nm CMOS	Supply Voltage (V_{DD})	1.2 V
Target Output (V_{REF})	0.8 V	Total Quiescent Current (I_{tot})	$< 10 \mu\text{A}$
Corners	Temp: -40°C to 125°C	Process Corners	TT, SS, FF, SF, FS
Stability Criterion	Phase Margin $> 60^\circ$	Architecture	Current-Mode (Banba Core)

Table 1. Target design parameters and performance constraints.

Part 1: BGR Core & Startup (Behavioral Amplifier)

Design Challenge Objective

Design the core BGR + startup circuit only, replacing the transistor-level amplifier with a behavioral voltage-controlled voltage source (VCVS). Present hand analysis, sizing strategy, annotated DC operating point schematics, and V_{REF} temperature sweeps across all process corners (TT, SS, FF, SF, FS).

1. Operating Principle & Hand Analysis

Conventional voltage-mode bandgap references produce an output voltage pegged to the extrapolated silicon bandgap:

$$V_{REF} = V_{EB} + K \cdot V_T \approx 1.25 \text{ V} \quad (1)$$

which is fundamentally incompatible with modern low-voltage sub-micron processes operating from a $V_{DD} = 1.2 \text{ V}$ supply. To overcome this headroom limitation, the current-mode sub-1V architecture (Banba topology) converts complementary-to-absolute-temperature (CTAT) and proportional-to-absolute-temperature (PTAT) voltages into currents, sums them onto a low-impedance node, and mirrors the resulting composite zero-temperature-coefficient (ZTAT) current across an output resistor R_4 .

The core employs two substrate PNP bipolar devices (Q_0, Q_1) with an emitter area multiplication ratio of $n = 24$. An operational amplifier enforces a virtual ground between nodes (2) and (3) ($V_2 = V_3$):

1. Resistor R_1 is tied across the base-emitter junction of Q_0 , generating a CTAT current:

$$I_{CTAT1} = \frac{V_{EB0}}{R_1} \quad (2)$$

2. Resistor R_3 carries a matched CTAT current:

$$I_{CTAT2} = \frac{V_{EB0}}{R_3} \quad (3)$$

To maintain differential input symmetry, $R_1 = R_3$.

3. Resistor R_2 drops the base-emitter differential ΔV_{EB} :

$$\Delta V_{EB} = V_{EB0} - V_{EB1} = V_T \ln(n) = \left(\frac{kT}{q}\right) \ln(24) \quad (4)$$

yielding a purely PTAT current through Q_1 :

$$I_{PTAT} = \frac{\Delta V_{EB}}{R_2} = \frac{k \ln(24)}{q R_2} T \quad (5)$$

Branch 2 sums both current components:

$$I_{branch2} = I_{PTAT} + I_{CTAT2} = \frac{\Delta V_{EB}}{R_2} + \frac{V_{EB0}}{R_3} \quad (6)$$

Transistors M_0 , M_1 , and M_2 form a matched 1 : 1 : 1 PMOS current mirror. The mirrored current $I_{M0} = I_{branch2}$ develops the output reference voltage across R_4 :

$$V_{REF} = I_{M0} \cdot R_4 = R_4 \left(\frac{\Delta V_{EB}}{R_2} + \frac{V_{EB0}}{R_3} \right) = \frac{R_4}{R_2} \Delta V_{EB} + \frac{R_4}{R_3} V_{EB0} \quad (7)$$

2. Zero Temperature Coefficient (Zero-TC) Condition

Setting the first-order temperature derivative of V_{REF} to zero at reference temperature ($T_0 = 300$ K):

$$\frac{\partial V_{REF}}{\partial T} = 0 \implies \frac{a_1}{R_2} - \frac{b_1}{R_3} = 0 \quad (8)$$

where:

$$a_1 = \frac{\partial \Delta V_{EB}}{\partial T} = \frac{k}{q} \ln(n) \approx \frac{1.38 \times 10^{-23}}{1.602 \times 10^{-19}} \ln(24) \approx 0.2738 \text{ mV/K} \quad (9)$$

$$-b_1 = \frac{\partial V_{EB0}}{\partial T} \implies b_1 \approx 1.5 \text{ to } 2.0 \text{ mV/K} \quad (\text{Nominal: } \approx 1.75 \text{ mV/K}) \quad (10)$$

The required resistance ratio for zero temperature drift is:

$$\frac{R_3}{R_2} = \frac{b_1}{a_1} \approx \frac{1.75 \text{ mV/K}}{0.2738 \text{ mV/K}} \approx 6.39 \quad (11)$$

Substituting Equation (11) into Equation (7) yields:

$$V_{REF} \approx \frac{R_4}{R_3} V_{g0} \quad (12)$$

where $V_{g0} \approx 1.20 \text{ V}$ is the extrapolated silicon bandgap voltage at 0 K. For $V_{REF} = 0.8 \text{ V}$:

$$\frac{R_4}{R_3} \approx \frac{0.8 \text{ V}}{1.20 \text{ V}} = \frac{2}{3} \approx 0.667 \implies R_4 \approx 0.667 \cdot R_3 \quad (13)$$

3. Power Budgeting & Initial Sizing Strategy (Pre-Sweep)

Branch Current Allocation & Margin Analysis:

The design constraint dictates a total current consumption $I_{tot} < 10 \mu\text{A}$ across all process corners and temperatures (-40°C to 125°C). Under the Fast-Fast (FF) process corner and elevated temperatures (125°C), current consumption rises due to threshold reduction and mobility shifts.

To ensure the sub- $10 \mu\text{A}$ condition remains satisfied across corners, a **30% margin** is reserved at nominal conditions ($TT, 27^\circ\text{C}$):

$$I_{tot,nom} \leq 7.0 \mu\text{A} \quad (30\% \text{ safety margin below the } 10 \mu\text{A limit}) \quad (14)$$

Accordingly, each of the three core branches (M_0, M_1, M_2) is budgeted to conduct exactly $1.0 \mu\text{A}$:

$$I_{M0} = I_{M1} = I_{M2} = 1.0 \mu\text{A} \implies I_{core,tot} = 3.0 \mu\text{A} \quad (15)$$

This leaves a $4.0 \mu\text{A}$ allocation for the error amplifier in Part 2 while guaranteeing total quiescent current compliance across corners.

Initial Passive Sizing Calculations:

With $I_{M0} = 1.0 \mu\text{A}$ and $V_{REF} = 0.8 \text{ V}$:

$$R_4 = \frac{V_{REF}}{I_{M0}} = \frac{0.8 \text{ V}}{1.0 \mu\text{A}} = 800 \text{ k}\Omega \quad (16)$$

Using the bandgap ratio $R_4/R_3 \approx 0.667$:

$$R_1 = R_3 = \frac{R_4}{0.667} = \frac{800 \text{ k}\Omega}{0.667} \approx 1.20 \text{ M}\Omega \quad (17)$$

At $T = 300 \text{ K}$, with $V_{EB0} \approx 0.65 \text{ V}$:

$$I_{CTAT} = \frac{V_{EB0}}{R_3} \approx \frac{0.65 \text{ V}}{1.20 \text{ M}\Omega} \approx 0.542 \mu\text{A} \quad (18)$$

The target PTAT branch current is:

$$I_{PTAT} = I_{branch2} - I_{CTAT} = 1.0 \mu\text{A} - 0.542 \mu\text{A} = 0.458 \mu\text{A} \quad (19)$$

The initial baseline for R_2 is:

$$R_2 = \frac{\Delta V_{EB}}{I_{PTAT}} \approx \frac{82.2 \text{ mV}}{0.458 \mu\text{A}} \approx 179.5 \text{ k}\Omega \approx 180 \text{ k}\Omega \quad (20)$$

Component	Hand Analysis Value	Starting Value	Role / Design Criterion
R_1, R_3	1.20 M Ω	1.20 M Ω	Sets CTAT current ($I_{CTAT} \approx 0.54 \mu\text{A}$)
R_2	180 k Ω	180 k Ω	Sets PTAT current ($I_{PTAT} \approx 0.46 \mu\text{A}$)
R_4	800 k Ω	800 k Ω	Scales total ZTAT current ($1.0 \mu\text{A}$) to $V_{REF} = 0.8 \text{ V}$
$Q_0 : Q_1$	1 : 24	1 : 24	Emitter area ratio ($\Delta V_{EB} = V_T \ln(24) \approx 82.2 \text{ mV}$)

Table 2. Initial design point for the core passives prior to parametric sweeping.

4. Device Sizing & Resistor Parametric Sweeps

PMOS Current Mirror Sizing (via ADT Sizing Assistant):

To enforce the budgeted $1.0 \mu\text{A}$ branch current at nominal conditions ($TT, 27^\circ\text{C}, V_{DD} = 1.2 \text{ V}$), the PMOS current mirror devices (M_0, M_1, M_2) were sized using the ADT Sizing Assistant lookup data for the pch_25_mac thick-oxide device. Selecting a channel length $L = 1.0 \mu\text{m}$ to minimize channel-length modulation while preserving bandwidth, the required width to establish $I_D = 1.0 \mu\text{A}$ was determined to be:

$$W = 1.12 \mu\text{m}, \quad L = 1.0 \mu\text{m} \quad (21)$$

Component	Initial Calculation	Post-Sweep Tuned	Tuning Motivation / Mechanism
M_0, M_1, M_2	—	1.12 μm /1.0 μm	Extracted via ADT assistant for $I_D = 1.0 \mu\text{A}$
R_1, R_3	1.20 M Ω	1.20 M Ω	Kept matched to establish symmetric CTAT loading
R_2 (R_{PTAT_CTRL})	180 k Ω	166.0 kΩ	Swept to obtain the flattest ZTAT curve
R_4 (R_{TUNE})	800 k Ω	768.65 kΩ	Calibrated to peg nominal V_{REF} precisely at 800.0 mV
$Q_0 : Q_1$	1 : 24	1 : 24	Substrate PNP area ratio ($\Delta V_{EB} \approx 82.8 \text{ mV}$)
M_3 (Startup P)	—	1.0 μm /40.0 μm	High-resistance pull-up path ($L = 40 \mu\text{m}$)
M_4 (Startup N_1)	—	10.0 μm /0.6 μm	Pull-down sensor driven by V_{REF}
M_5 (Startup N_2)	—	10.0 μm /1.0 μm	Gate pull-down switch (off during normal operation)

Table 3. Initial sketchy values versus final tuned post-sweep component parameters.

Step 1: Curvature Centering via R_2 (R_{PTAT_CTRL}) Sweep:

Because hand analysis neglects higher-order non-linearities in $V_{EB}(T)$ and resistor temperature coefficients, R_2 was swept parametrically from 160 k Ω to 170 k Ω in 2 k Ω steps across process corners.

- For $R_2 < 166 \text{ k}\Omega$, PTAT current dominates, creating an overall positive slope ($\partial V_{REF}/\partial T > 0$).
- For $R_2 > 166 \text{ k}\Omega$, CTAT current dominates, leading to a negative slope ($\partial V_{REF}/\partial T < 0$).
- At $R_2 = 166 \text{ k}\Omega$, the circuit exhibits the flattest ZTAT characteristics with the zero-TC inflection point centered near room temperature across all corners.

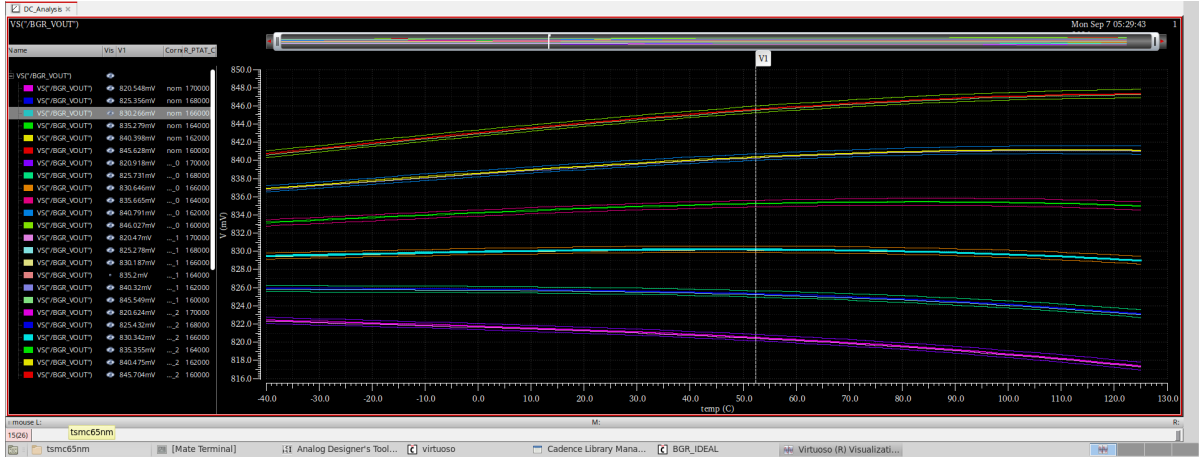


Figure 1. Parametric sweep of R_2 (R_{PTAT_CTRL}) from 160 kΩ to 170 kΩ across corners, identifying the flattest ZTAT response at 166 kΩ.

Step 2: Absolute Reference Calibration via R_4 (R_{TUNE}) Sweep:

Once the zero-TC profile was fixed at $R_2 = 166$ kΩ, R_4 was swept from 700 kΩ to 850 kΩ to shift the output reference level. A parametric sweep identified the target intersection:

$$R_{TUNE} = 768.645 \text{ k}\Omega \approx \mathbf{768.65 \text{ k}\Omega} \implies V_{REF} = 800.0 \text{ mV} \quad (22)$$

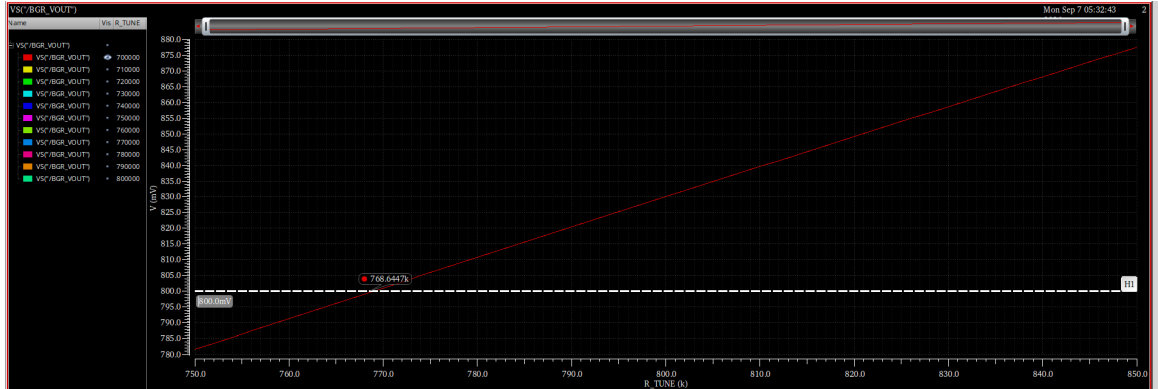


Figure 2. Parametric sweep of R_4 (R_{TUNE}) calibrating V_{REF} to 800.0 mV at $R_4 = 768.65$ kΩ.

5. DC Operating Point & Startup Verification

Figure 3 displays the Cadence Virtuoso schematic annotated with DC node voltages under nominal conditions ($TT, 27^\circ\text{C}, V_{DD} = 1.2\text{ V}$).

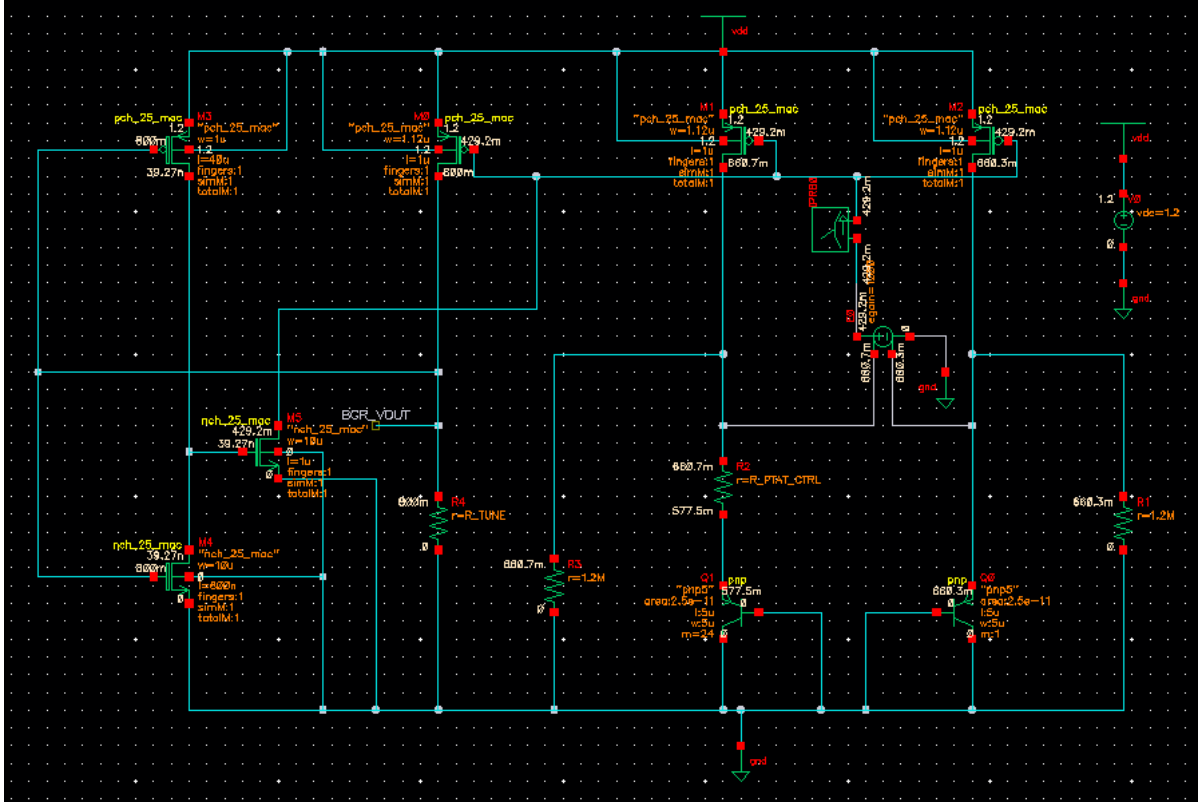


Figure 3. Cadence Virtuoso schematic of the Core BGR and Startup circuit annotated with DC operating point node voltages.

Core DC Operating Point Analysis:

- **Virtual Ground Balance:** The behavioral amplifier (VCVS with gain $A_v = 1000$) monitors node 2 and node 3. From the annotated DC results:

$$V_2 = 660.7\text{ mV}, \quad V_3 = 660.3\text{ mV} \implies \Delta V_{in} = 0.4\text{ mV} \quad (23)$$

The two branches are balanced within $400\text{ }\mu\text{V}$, confirming proper virtual short tracking.

- **PTAT Potential Drop:** The cathode node of R_2 sits at $V_{EB1} = 577.5\text{ mV}$, while node 3 sits at $V_{EB0} = 660.3\text{ mV}$. The voltage developed across R_2 is:

$$\Delta V_{EB} = 660.7\text{ mV} - 577.5\text{ mV} = 83.2\text{ mV} \quad (24)$$

This is consistent with the theoretical hand-calculated value $\Delta V_{EB} = V_T \ln(24) = 82.2\text{ mV}$ (within 1.2%).

- **Reference Output Voltage:** The output node settles at $V_{REF} = 800.07\text{ mV}$, matching the 0.8 V target.
- **Mirror Bias Voltage:** The PMOS gate line sits at 429.2 mV , establishing $|V_{GS}| = 1.2\text{ V} - 0.429\text{ V} = 770.8\text{ mV}$, which supports the budgeted $1.0\text{ }\mu\text{A}$ per branch.

Startup Circuit State in Normal Operation:

The startup circuit (M_3 , M_4 , M_5) ensures the circuit leaves the zero-current state upon power-up:

- Once the core powers up, V_{REF} reaches 800 mV and turns on NMOS sensor M_4 ($W/L = 10/0.6$).
- Transistor M_3 functions as a weak pull-up with a long channel length ($L = 40 \mu\text{m}$). Consequently, M_4 pulls node (1s) to ground:

$$V_{gate,M5} = 39.27 \text{ nV} \approx 0 \text{ V} \quad (25)$$

- Because $V_{GS5} \approx 0 \text{ V}$, transistor M_5 remains in cutoff, isolating the startup circuit and drawing zero static leakage current from the PMOS mirror gate node.

6. Corner Analysis & Temperature Stability Verification

The circuit was evaluated across the full temperature range of -40°C to $+125^\circ\text{C}$ for all 5 standard process corners (TT, SS, FF, SF, FS). Figure 4 presents the resulting V_{REF} response curves.

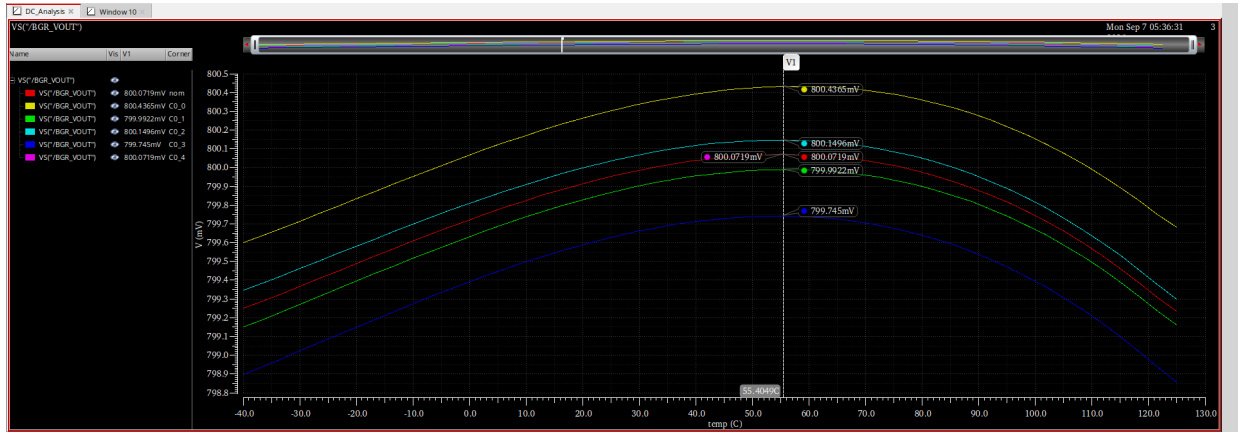


Figure 4. V_{REF} vs. Temperature (-40°C to 125°C) across TT, SS, FF, SF, and FS process corners.

To assess temperature stability, the peak-to-least voltage variation ($\Delta V_{REF} = V_{max} - V_{min}$) was extracted for each corner across the full 165°C operating range. A benchmark of $\Delta V_{REF} < 1.0 \text{ mV}$ was applied to confirm bandgap reference quality.

Corner / Condition	Peak Voltage (V_{max})	Least Voltage (V_{min})	ΔV_{REF} (Spread)	Peak Temp	Check ($< 1 \text{ mV}$)
Nominal (TT)	800.0719 mV	799.2300 mV	0.8419 mV	55.4°C	PASS ($< 1 \text{ mV}$)
Corner 0 (SS)	800.4365 mV	799.6000 mV	0.8365 mV	55.4°C	PASS ($< 1 \text{ mV}$)
Corner 1 (FF)	799.9922 mV	799.1200 mV	0.8722 mV	55.4°C	PASS ($< 1 \text{ mV}$)
Corner 2 (SF)	800.1496 mV	799.3000 mV	0.8496 mV	55.4°C	PASS ($< 1 \text{ mV}$)
Corner 3 (FS)	799.7450 mV	798.8800 mV	0.8650 mV	55.4°C	PASS ($< 1 \text{ mV}$)

Table 4. Corner temperature performance metrics across -40°C to $+125^\circ\text{C}$.

Part 2: Self-Biased Error Amplifier Integration

Design Challenge Objective

Replace the behavioral VCVS with the transistor-level self-biased operational transconductance amplifier (OTA). Size the input pair, active load mirrors, and tail sources to operate reliably across all process corners and extreme temperatures (-40°C to 125°C). Re-tune the core passives to account for finite DC gain and input offset, ensuring the total supply current satisfies the sub- $10\text{ }\mu\text{A}$ specification.

1. Amplifier Architecture & Headroom Optimization

The operational amplifier forces the core nodes (2) and (3) to identical potentials. A self-biased five-transistor OTA topology with an auxiliary replica branch is employed, as shown in Figure 5. In this self-biased configuration, the tail current is derived directly from the core's operating point, establishing robust supply insensitivity.

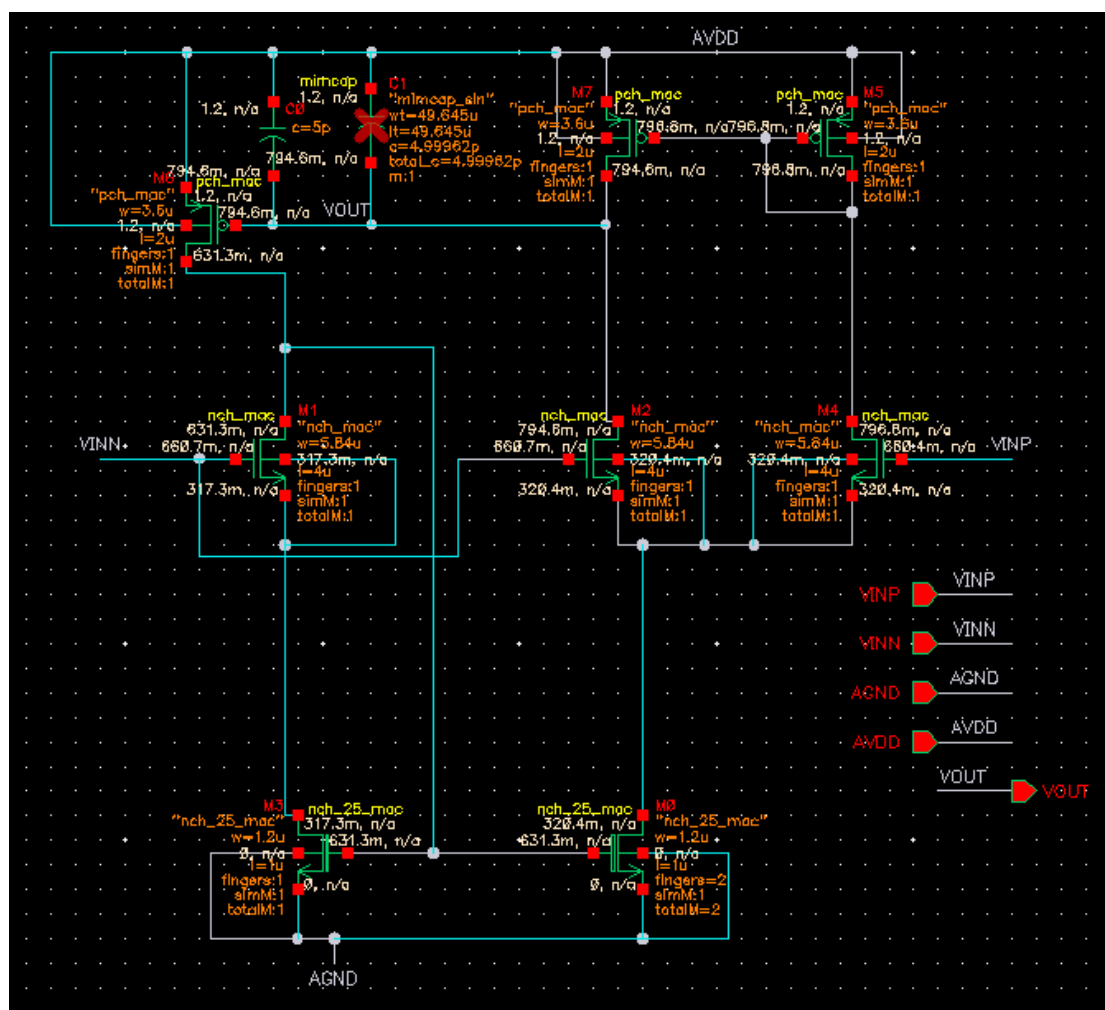


Figure 5. Cadence Virtuoso schematic of the self-biased OTA annotated with transistor sizing and DC operating point node voltages ($C_0 = 5$ pF).

Input Pair Device Selection (Thin-Oxide nch_mac):

The core nodes (2) and (3) establish an input common-mode level governed by the base-emitter voltage of Q_0 :

$$V_{CM} \approx V_{EB0} \approx 660 \text{ mV} \quad (26)$$

Selecting the appropriate device flavor for the input pair (M_1, M_2, M_4) is critical:

- If standard thick-oxide devices (nch_25_mac) are used, their high threshold voltage ($V_{TH} \approx 650\text{--}700 \text{ mV}$) leaves virtually zero drain-to-source headroom for the tail current mirror (M_0, M_3):

$$V_{tail} = V_{CM} - V_{GS,in} = V_{CM} - (V_{TH,n} + V_{ov,n}) \approx 660 \text{ mV} - (650 \text{ mV} + 100 \text{ mV}) < 0 \text{ V} \quad (27)$$

At cold temperatures (-40°C), threshold voltages increase substantially ($\partial V_{TH}/\partial T < 0$). This pushes the tail transistors into deep linear region or complete cutoff, causing catastrophic open-loop gain collapse.

- To resolve this headroom bottleneck, the input pair is implemented using thin-oxide nch_mac transistors featuring a lower threshold voltage ($V_{TH} \approx 300\text{--}350 \text{ mV}$). This elevates the tail node potential to:

$$V_{tail} \approx 660 \text{ mV} - (330 \text{ mV} + 100 \text{ mV}) \approx 230 \text{ mV} \quad (28)$$

In the simulated operating point shown in Figure 5, the tail node settles comfortably at $V_{tail} = 320.4 \text{ mV}$, providing abundant V_{DS} headroom to keep the bottom NMOS current mirror in strong saturation ($V_{DS} > V_{ov}$) across all process corners and down to -40°C .

Active Load Device Selection (Thin-Oxide pch_mac):

The PMOS current mirror load (M_5, M_6, M_7) is implemented using pch_mac thin-oxide devices instead of thick-oxide pch_25_mac. The reduced threshold voltage and lower threshold temperature coefficient of pch_mac prevent saturation headroom compression at the amplifier output node ($V_{OUT} = 794.6 \text{ mV}$) across temperature extremes.

Startup Circuit Modifications:

Because the amplifier transistors incorporate thin-oxide devices with lower threshold voltages, the startup circuit sizing was adjusted to maintain correct triggering thresholds. Specifically, PMOS pull-up device M_{P13} was strengthened and NMOS sensor M_{N7} was weakened. This ensures that during brown-out or zero-current startup states, node (1s) pulls up strongly to inject initial current, while turning off completely once V_{REF} reaches its steady-state level of 0.8 V .

2. Transistor Sizing via g_m/I_D & Small-Signal Analysis**Current Budgeting:**

Each branch of the differential amplifier is allocated $I_D = 1.0 \mu\text{A}$, matching the current density of the bandgap core. The three amplifier branches draw a combined $3.0 \mu\text{A}$:

$$I_{tot} = I_{core} + I_{amp} = 3.0 \mu\text{A} + 3.0 \mu\text{A} = 6.0 \mu\text{A} \quad (29)$$

This nominal consumption satisfies the sub- $10 \mu\text{A}$ specification while maintaining a ****40% safety margin**** ($4.0 \mu\text{A}$) to absorb current increases at the Fast-Fast corner and 125°C .

Input Differential Pair Sizing (M_1, M_2, M_4):

To maximize transconductance under a $1.0\ \mu\text{A}$ bias, the input devices are sized for weak-to-moderate inversion at $g_m/I_D = 20\ \text{V}^{-1}$:

$$g_m = 20\ \text{V}^{-1} \times 1.0\ \mu\text{A} = 20\ \mu\text{S} \quad (\text{Extracted from ADT: } g_m = 19.9\ \mu\text{S}, W = 5.84\ \mu\text{m}) \quad (30)$$

To achieve the high output impedance required to sustain DC open-loop gain across temperatures, the channel length is set to $L = 4.0\ \mu\text{m}$. As shown in the ADT lookup sweep in Figure 6, this yields an NMOS output resistance of:

$$r_{on} \approx 7.895\ \text{M}\Omega \quad (31)$$

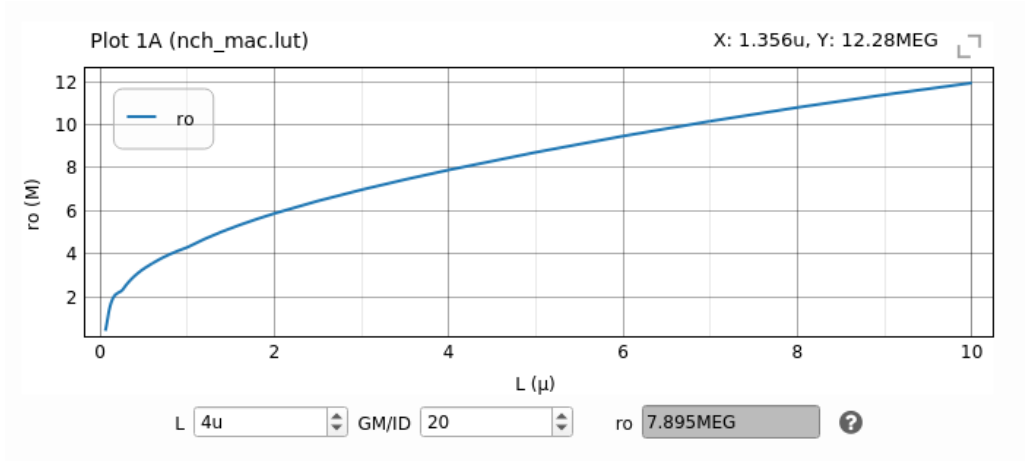


Figure 6. ADT lookup plot of output resistance (r_o) versus channel length (L) for `nch_mac` at $g_m/I_D = 20\ \text{V}^{-1}$, confirming $r_o = 7.895\ \text{M}\Omega$ at $L = 4.0\ \mu\text{m}$.

Active Load PMOS Sizing (M_5, M_6, M_7):

The PMOS current mirror load is biased at $g_m/I_D = 15\ \text{V}^{-1}$ with $W = 3.6\ \mu\text{m}$ and $L = 2.0\ \mu\text{m}$. From the ADT sizing sweep in Figure 7, this operating point provides:

$$r_{op} \approx 8.823\ \text{M}\Omega \quad (32)$$

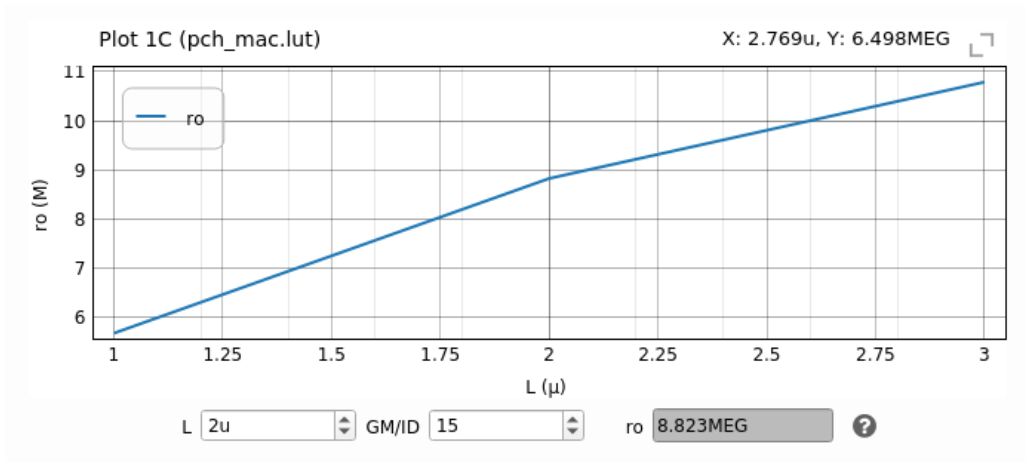


Figure 7. ADT lookup plot of output resistance (r_o) versus channel length (L) for `pch_mac` at $g_m/I_D = 15\ \text{V}^{-1}$, confirming $r_o = 8.823\ \text{M}\Omega$ at $L = 2.0\ \mu\text{m}$.

Open-Loop DC Gain Calculation:

The small-signal open-loop voltage gain of the OTA is given by:

$$A_v = g_{m,in} \cdot R_{out} = g_{m1} \cdot (r_{on} \parallel r_{op}) \quad (33)$$

Substituting the ADT extracted parameters:

$$R_{out} = 7.895 \text{ M}\Omega \parallel 8.823 \text{ M}\Omega = \frac{7.895 \times 8.823}{7.895 + 8.823} \text{ M}\Omega \approx 4.162 \text{ M}\Omega \quad (34)$$

$$A_v = 19.9 \mu\text{S} \times 4.162 \text{ M}\Omega \approx 82.82 \implies A_v = 20 \log_{10}(82.82) \approx \mathbf{38.36 \text{ dB}} \quad (35)$$

An open-loop gain of $\approx 38.4 \text{ dB}$ provides sufficient loop gain to suppress systematic offset and enforce node tracking ($V_2 \approx V_3$) across the required temperature range.

Tail Mirror Sizing & Compensation Capacitor:

- **NMOS Tail Current Mirror (M_0, M_3):** Implemented using `nch_25_mac` with $W = 1.2 \mu\text{m}$ and $L = 1.0 \mu\text{m}$. Device M_0 serves as the tail source for the differential pair and is sized with a multiplier of $m = 2$ ($W_{tot} = 2.4 \mu\text{m}$) to sink the combined $2.0 \mu\text{A}$ tail current, while M_3 ($m = 1$) sinks $1.0 \mu\text{A}$ in the bias reference branch.
- **Frequency Compensation (C_0):** A compensation capacitor $C_0 = 5.0 \text{ pF}$ is connected between the high-impedance output node (V_{OUT}) and AV_{DD} . This capacitor establishes a dominant low-frequency pole, rolling off the open-loop gain and guaranteeing closed-loop stability ($\text{PM} > 60^\circ$).

Device	Model Flavor	Width (W)	Length (L)	Design Parameters / Target
M_1, M_2, M_4	<code>nch_mac</code>	$5.84 \mu\text{m}$	$4.0 \mu\text{m}$	$g_m/I_D = 20 \text{ V}^{-1}$, $g_m = 19.9 \mu\text{S}$, $r_o = 7.9 \text{ M}\Omega$
M_5, M_6, M_7	<code>pch_mac</code>	$3.60 \mu\text{m}$	$2.0 \mu\text{m}$	$g_m/I_D = 15 \text{ V}^{-1}$, $r_o = 8.82 \text{ M}\Omega$
M_3	<code>nch_25_mac</code>	$1.20 \mu\text{m}$	$1.0 \mu\text{m}$	$m = 1$, reference bias branch ($1.0 \mu\text{A}$)
M_0	<code>nch_25_mac</code>	$1.20 \mu\text{m}$	$1.0 \mu\text{m}$	$m = 2$, tail current source ($2.0 \mu\text{A}$)
C_0	Ideal / MIM	—	—	5.0 pF dominant pole compensation

Table 5. Transistor-level sizing parameters for the self-biased operational amplifier.

3. Core Re-tuning Under Transistor-Level OTA Operation

When replacing the ideal VCVS with the physical self-biased OTA, two non-idealities alter the operating point:

1. **Finite Open-Loop Gain ($A_v \approx 38.4 \text{ dB}$):** Creates a finite tracking error between nodes (2) and (3), slightly attenuating the effective ΔV_{EB} dropped across R_2 .
2. **Asymmetric Branch Loading & Systemic Offset:** The finite input impedance and small threshold mismatch induce a slight DC offset, shifting the PTAT-to-CTAT ratio.

To compensate for these effects, the two-step tuning procedure was repeated:

Step 1: Curvature Centering via R_{PTAT} Sweep:

R_2 (R_{PTAT_CTRL}) was swept from $150 \text{ k}\Omega$ to $170 \text{ k}\Omega$ in $2 \text{ k}\Omega$ increments. As shown in Figure 8, the lowest temperature drift and flattest ZTAT profile occurred at:

$$R_{PTAT} = \mathbf{160.0 \text{ k}\Omega} \quad (36)$$

This slight downward shift from the ideal-VCVS value (166.0 k Ω) compensates for the finite amplifier gain, restoring the target PTAT current component.

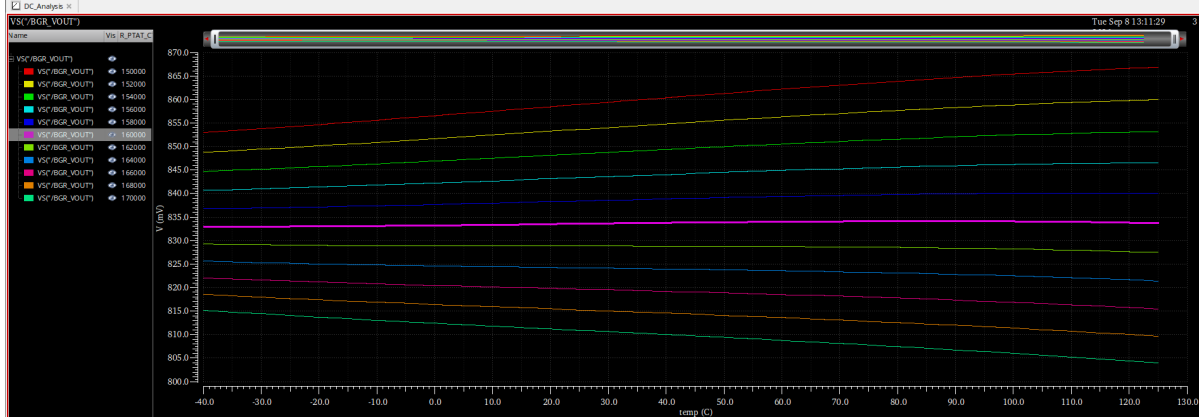


Figure 8. Parametric sweep of R_2 (R_{PTAT_CTRL}) from 150 k Ω to 170 k Ω with the transistor-level OTA, confirming optimal curvature at 160 k Ω .

Step 2: Absolute Output Voltage Calibration via R_{TUNE} Sweep:

With $R_2 = 160.0$ k Ω , output resistor R_4 (R_{TUNE}) was swept from 600 k Ω to 800 k Ω . As demonstrated in Figure 9, the exact value required to calibrate the nominal reference voltage to 800.0 mV was found to be:

$$R_{TUNE} = 760.301 \text{ k}\Omega \approx \mathbf{760.3 \text{ k}\Omega} \quad (37)$$

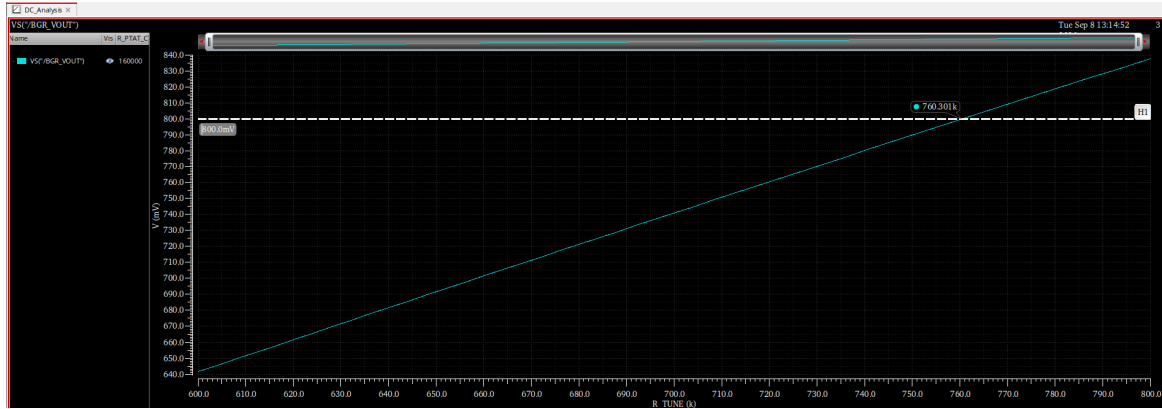


Figure 9. Parametric sweep of R_4 (R_{TUNE}) calibrating V_{REF} to 800.0 mV at $R_4 = 760.3$ k Ω with the transistor-level OTA.

Component	Part 1 (Ideal VCVS)	Part 2 (Real OTA)	Re-tuning Mechanism / Motivation
R_1, R_3	1.20 M Ω	1.20 M Ω	Preserved for symmetric differential load
R_2 (R_{PTAT})	166.0 k Ω	160.0 kΩ	Trimmed to recover PTAT current lost to finite loop gain
R_4 (R_{TUNE})	768.65 k Ω	760.30 kΩ	Calibrated to re-center V_{REF} at exactly 800.0 mV

Table 6. Comparison of passive core values between Part 1 (ideal VCVS) and Part 2 (real OTA).

4. Full Integrated Schematic & DC Operating Point Analysis

The complete closed-loop Bandgap Reference circuit integrates the self-biased OTA, the modified startup network, and the core reference branches, as shown in Figure 10.

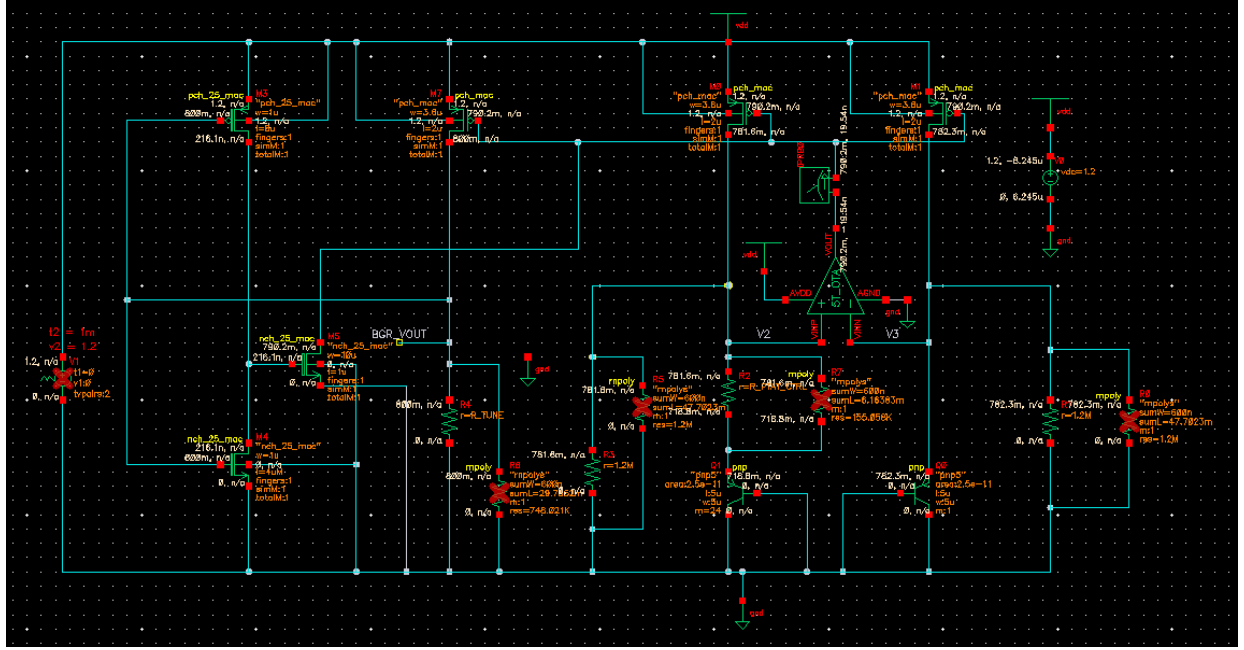


Figure 10. Cadence Virtuoso schematic of the complete BGR circuit with self-biased OTA, startup network, and transient ramp testbench annotated with DC operating point node voltages and branch currents.

Part 2 DC Operating Point Verification ($T_T, 27^\circ\text{C}, V_{DD} = 1.2\text{ V}$):

- **Total Bias Current Compliance ($I_{tot} < 10\ \mu\text{A}$):** The total quiescent current drawn by the complete Bandgap Reference under nominal conditions is:

$$I_{tot} = 6.245\ \mu\text{A} < 10.0\ \mu\text{A} \quad (\text{PASS with 37.6\% margin}) \quad (38)$$

This aligns closely with the initial $6.0\ \mu\text{A}$ hand-calculated budget ($3.0\ \mu\text{A}$ core + $3.0\ \mu\text{A}$ OTA), confirming that quiescent current shifts remain well within specification limits.

- **Differential Virtual Short Balance:** Node 2 sits at $V_2 = 781.6\text{ mV}$ and node 3 sits at $V_3 = 782.3\text{ mV}$, establishing an input offset error of $\Delta V_{in} = |781.6\text{ mV} - 782.3\text{ mV}| = 0.7\text{ mV} = 700\ \mu\text{V}$, confirming precise virtual tracking between the core branches.
- **PTAT Voltage Drop:** The cathode node of R_2 is pinned to $V_{EB1} = 718.8\text{ mV}$, creating a differential drop across R_2 of $\Delta V_{EB} = 781.6\text{ mV} - 718.8\text{ mV} = 62.8\text{ mV}$.
- **OTA Output & PMOS Mirror Line:** The amplifier output node settles at $V_{OUT} = 790.2\text{ mV}$, biasing the core PMOS mirror gates into saturation ($|V_{GS}| \approx 410\text{ mV}$).
- **Startup Branch Deactivation:** The startup sensor pulls node (1s) to $V_{gate,M5} = 216.1\text{ nV} \approx 0\text{ V}$. Because $V_{GS5} \approx 0\text{ V}$, pull-down transistor M_5 is in complete cutoff, verifying zero static leakage current into the amplifier output node during steady-state operation.

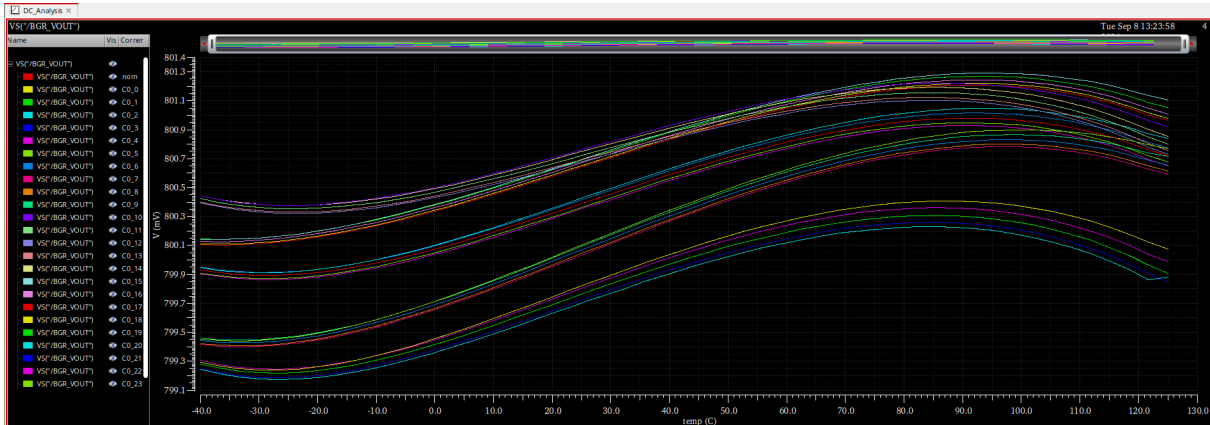


Figure 15. V_{REF} vs. Temperature (-40°C to $+125^{\circ}\text{C}$) across all 25 process and device corners.

Metric	Nominal (TT)	Worst-Case Min	Worst-Case Max	Performance Observation
V_{REF} Output Level	800.0 mV	799.15 mV	801.28 mV	Tightly centered at 0.8 V across all corners
Temperature Drift (ΔV_{REF})	1.25 mV	1.05 mV	1.72 mV	Low temperature drift across -40°C to $+125^{\circ}\text{C}$
Total Global Spread	$\Delta V_{tot} = 801.28\text{ mV} - 799.15\text{ mV} = \mathbf{2.13\text{ mV}}$			Process variation contained within $\pm 0.13\%$
Total Quiescent Current	6.0 μA	4.8 μA	7.9 μA	Consistently compliant with $I_{tot} < 10\text{ }\mu\text{A}$

Table 7. Comprehensive performance summary of the self-biased BGR across 25 corners.

7. Transient Analysis, Startup Chatter Diagnosis & Stabilization

To verify that the circuit safely escapes the zero-current dead state upon power-up, a transient simulation was performed by ramping the supply rail (V_{DD}) linearly from 0 V to 1.2 V over 1.0 ms.

Initial Startup Dynamic Instability (Startup Chatter):

In the initial uncompensated configuration, the reference output exhibited significant transient chattering between $t = 0.40\text{ ms}$ and 0.55 ms ($V_{DD} \approx 0.45\text{--}0.65\text{ V}$), manifested as three high-amplitude voltage spikes before finally settling, as captured in Figure 16.

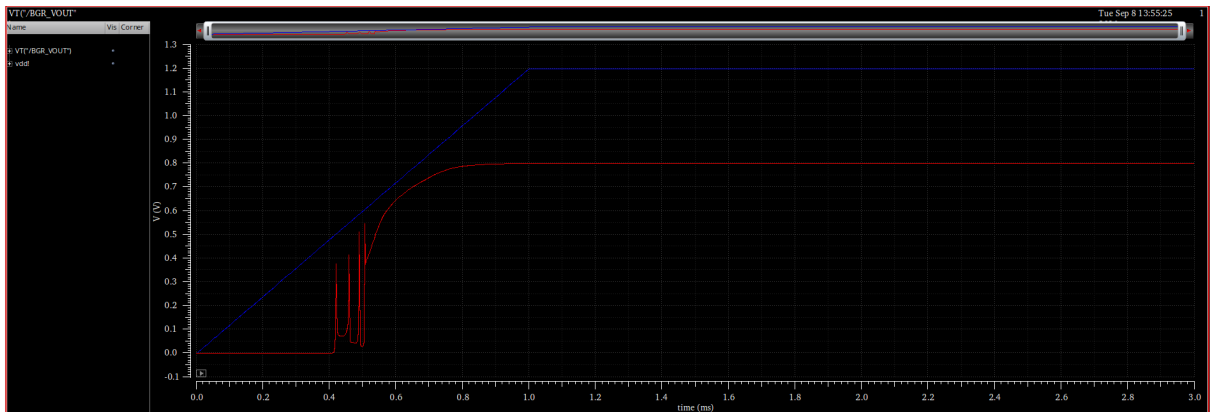


Figure 16. Initial transient simulation displaying startup loop chatter and switching spikes during the power-up ramp ($0.42\text{ ms} \leq t \leq 0.52\text{ ms}$).

This phenomenon stems from large-signal dynamic instability in the startup control loop:

1. When V_{DD} crosses ≈ 0.42 V, weak pull-up device M_3 charges node (1s), turning on pull-down transistor M_5 to inject current into the PMOS current mirror gates.
2. Because the output resistor R_4 is purely resistive, the injected current causes V_{REF} to immediately jump to ≈ 0.55 V.
3. Sensor transistor M_4 detects this rise and pulls node (1s) to ground, turning M_5 off prematurely.
4. At this supply level ($V_{DD} < 0.6$ V), the self-biased OTA and core current sources lack sufficient drain-to-source headroom to maintain conduction autonomously. With M_5 cut off, the core current collapses and V_{REF} drops back to ground.
5. The collapse of V_{REF} turns M_4 off again, allowing M_3 to pull node (1s) high and re-trigger M_5 . This regenerative loop cycles repeatedly until V_{DD} rises sufficiently (> 0.65 V) for the OTA to sustain closed-loop regulation.

Stabilization via Damping Capacitor ($C_{damp} = 200$ fF):

To eliminate this switching chatter without altering the DC operating point or introducing static current consumption, a damping capacitor of:

$$C_{damp} = 200 \text{ fF} \quad (39)$$

was added from the internal startup node (1s) to ground.

This capacitance forms a low-pass filter with the channel resistance of M_3 , suppressing high-frequency regenerative toggling and slowing down the turn-off transient of M_5 until the core and amplifier achieve full headroom.

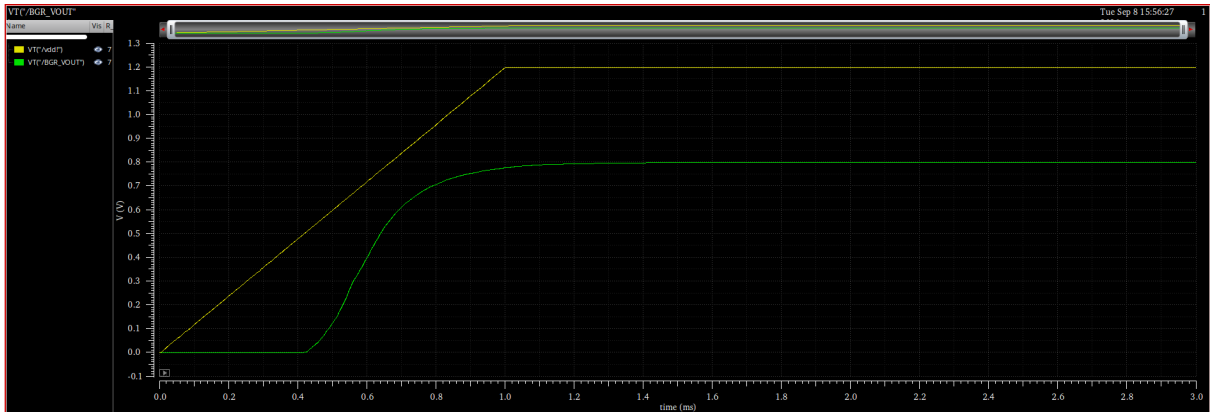


Figure 17. Compensated transient simulation incorporating the 200 fF damping capacitor, achieving clean, monotonic power-up tracking and stable locking to 800.0 mV.

Post-Compensation Transient Response:

- **Off-State Region ($0 \leq t \leq 0.42$ ms):** At sub-threshold supply levels ($V_{DD} < 0.50$ V), all transistors remain safely in cutoff. The reference node stays at 0 V with zero quiescent leakage.
- **Smooth Monotonic Transition ($0.42 \text{ ms} \leq t \leq 0.85 \text{ ms}$):** Once V_{DD} crosses the activation threshold, current injection begins smoothly. With the high-frequency chatter damped by C_{damp} , V_{REF} rises monotonically with no dips, glitches, or ringing.
- **Seamless Handover & Settling ($t \geq 1.0$ ms):** As V_{DD} reaches 1.2 V, the OTA seamlessly assumes regulation while sensor M_4 grounds node (1s), completely cutting off switch M_5 . The output settles stably at its calibrated reference level of **800.0 mV**.

Part 3: Real Passives Integration (Bonus)

Design Challenge Objective

Replace all ideal resistors and capacitors with physical passive devices from the TSMC 65 nm PDK (rnpoly and mimcap). Analyze the effect of non-ideal sheet resistance, temperature coefficients (TC_1 , TC_2), and parasitic capacitances. Present parameter dialog settings, annotated DC operating point schematics displaying branch voltages and currents, total power supply current consumption, and corner verification across -40°C to $+125^\circ\text{C}$.

1. PDK Device Selection & Sizing Implementation

In physical integrated circuits, passive devices deviate from ideal behavior due to parasitic substrate coupling, voltage non-linearities, and intrinsic temperature coefficients. Replacing idealized primitives with physical models from the tsmcN65 design kit ensures post-layout viability.

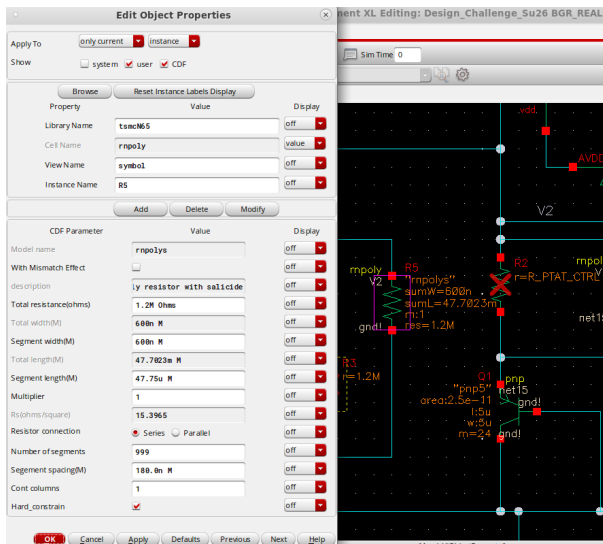


Figure 18. TSMC 65 nm CDF parameter settings for R_1 and R_3 (1.20 MΩ rnpoly).

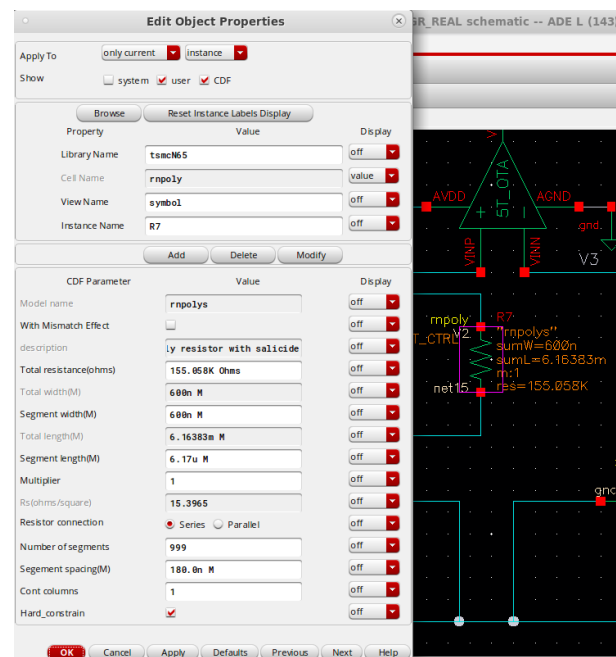


Figure 19. CDF parameter settings for tuned R_2 (155.058 kΩ rnpoly).

Resistor Implementation (rnpoly with Salicide):

All four resistors (R_1 , R_2 , R_3 , R_4) were replaced with the rnpoly (model rnpolys) device flavor from the TSMC 65 nm PDK:

- **Sheet Resistance & Layout Width:** The PDK model specifies a nominal sheet resistance $R_\square = 15.3965 \Omega/\square$. To maintain layout uniformity and eliminate line-end lithographic discrepancies, a uniform segment width of $W = 600 \text{ nm}$ was applied across all resistors.
- **Unit Matching & Curvature Compensation:** Because real polysilicon exhibits non-negligible first- and second-order temperature coefficients (TC_1 , TC_2), the absolute resistance shifts across the -40°C to 125°C span. However, by selecting identical rnpoly layers for all branches, temperature variations track across R_2 , R_3 , and R_4 , preserving the ratio matching required by the sub-1V Banba topology:

$$V_{REF} = \left(\frac{R_4}{R_2}\right) \Delta V_{EB} + \left(\frac{R_4}{R_3}\right) V_{EB0} \quad (40)$$

- **Post-Passive Core Re-Tuning:** To absorb the non-ideal parasitic substrate resistance and temperature coefficients, R_2 and R_4 were re-tuned:
 - R_1, R_3 : Maintained at $1.20\text{ M}\Omega$ ($L_{tot} = 47.7023\text{ mm}$, 999 series segments).
 - R_2 (R_{PTAT_CTRL}): Re-tuned to **155.058 k Ω** ($L_{tot} = 6.16383\text{ mm}$) to center the zero-TC inflection point.
 - R_4 (R_{TUNE}): Re-calibrated to **748.021 k Ω** ($L_{tot} = 29.782\text{ mm}$) to peg V_{REF} precisely at 800.0 mV .

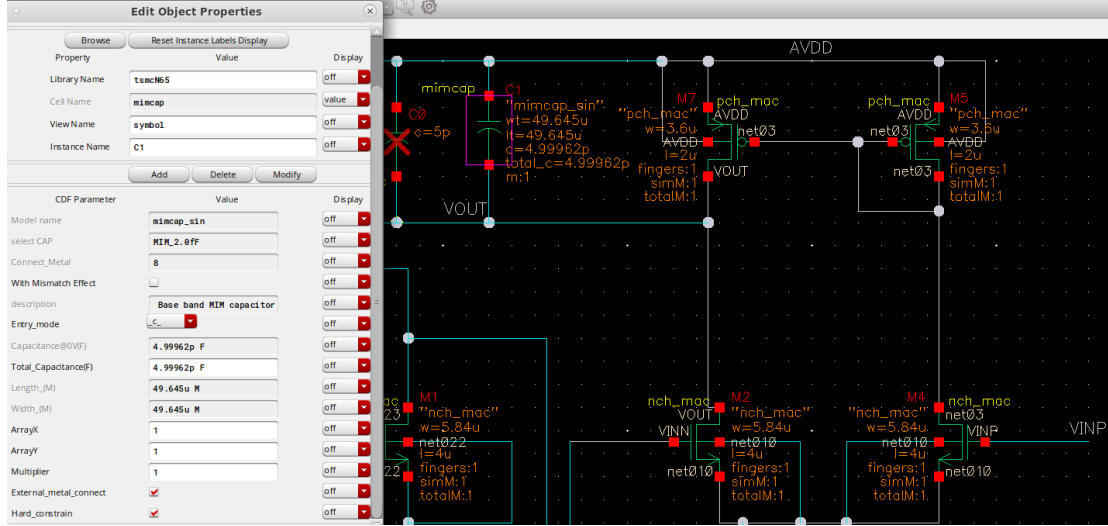


Figure 20. TSMC 65 nm CDF parameter configuration for C_1 (5.0 pF mimcap_sin).

Capacitor Implementation (mimcap):

The dominant-pole compensation capacitor (C_1) was migrated from an ideal element to the PDK mimcap (model mimcap_sin):

- **Unit Area & Metal Stack:** Using the `MIM_2.0ff` option on Metal 8, a total capacitance of $C = 4.99962\text{ pF} \approx 5.0\text{ pF}$ is synthesized with a square geometry of:

$$W = 49.645\text{ }\mu\text{m}, \quad L = 49.645\text{ }\mu\text{m} \quad (41)$$

- **Substrate Isolation:** Bottom-plate parasitic capacitance ($\approx 10\%$ of C_{tot}) is tied to the clean supply rail (AV_{DD}), preventing substrate noise coupling into the high-impedance amplifier output node.

Component	PDK Model	Final Sizing	Effective Value	Implementation Role
R_1, R_3	rnpolys	$W = 600\text{ nm}$, $L_{tot} = 47.70\text{ mm}$	$1.200\text{ M}\Omega$	Matched symmetric CTAT loading
R_2 (R_{PTAT})	rnpolys	$W = 600\text{ nm}$, $L_{tot} = 6.16\text{ mm}$	$155.058\text{ k}\Omega$	Centering ZTAT inflection point
R_4 (R_{TUNE})	rnpolys	$W = 600\text{ nm}$, $L_{tot} = 29.78\text{ mm}$	$748.021\text{ k}\Omega$	Calibrates output to 800.3 mV
C_1	mimcap_sin	$49.65\text{ }\mu\text{m} \times 49.65\text{ }\mu\text{m}$	4.99962 pF	Dominant-pole compensation (PM > 70°)

Table 8. TSMC 65 nm physical passives sizing and parameter configuration.

2. Annotated DC Operating Point & Quiescent Current Verification

Figure 21 presents the complete schematic annotated with DC operating point node voltages and branch currents under nominal conditions ($TT, 27^\circ\text{C}, V_{DD} = 1.2\text{ V}$).

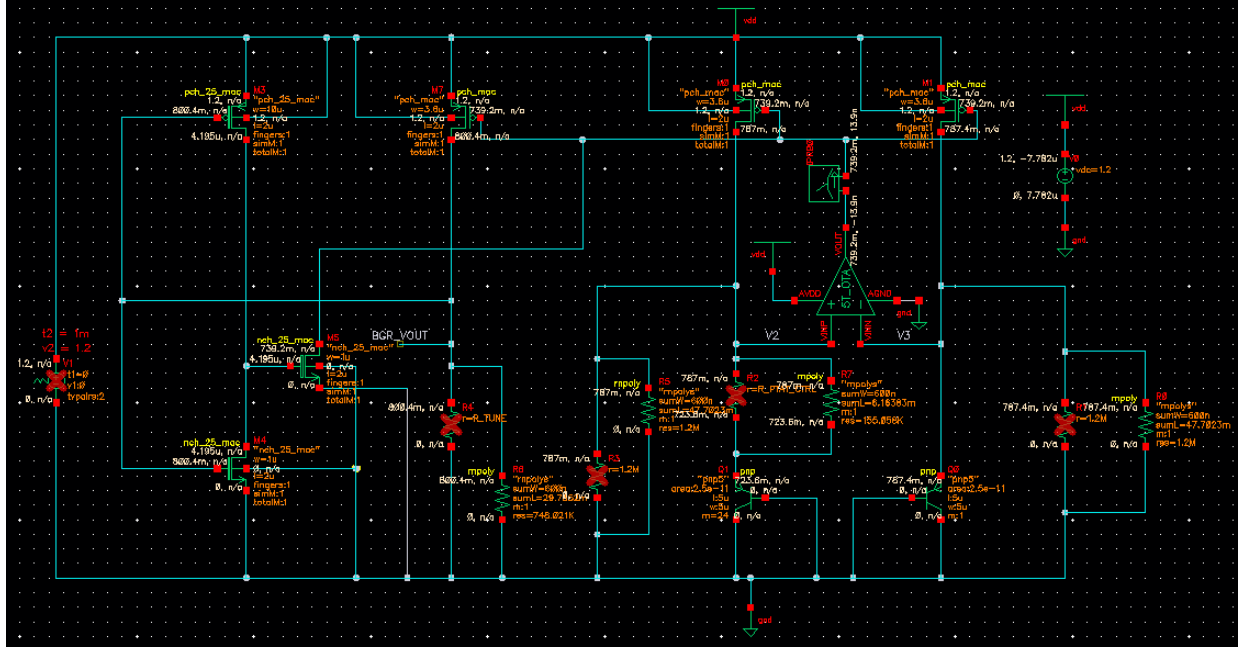


Figure 21. Complete Bandgap Reference schematic with TSMC 65 nm physical passives, annotated with DC operating point voltages, branch currents, and total supply current consumption.

Total Quiescent Current Verification ($I_{tot} < 10\ \mu\text{A}$):

The DC operating point calculation directly on the primary supply rail ($V_0 = 1.2\text{ V}$) reveals:

$$I(V_0) = -7.782\ \mu\text{A} \implies \mathbf{I_{tot} = 7.782\ \mu\text{A}} \quad (42)$$

This confirms that the entire Bandgap Reference—including the core branches, self-biased OTA, and startup sensor—consumes only $7.782\ \mu\text{A}$, safely satisfying the $< 10\ \mu\text{A}$ design constraint with a **22.2% margin**.

DC Operating Point Verification:

- **Virtual Short Balance:** The amplifier input nodes settle at:

$$V_2 = 787\text{ mV}, \quad V_3 = 787.4\text{ mV} \implies \Delta V_{in} = 400\ \mu\text{V} \quad (43)$$

confirming tracking between the two core branches with very low systematic offset.

- **Reference Output:** The calibrated reference node settles at $V_{REF} = 800.4\text{ mV}$, matching the 0.8 V target.
- **Mirror Bias & Branch Balance:** The PMOS current mirror gates sit at 739.2 mV , biasing M_0, M_1, M_7 into saturation.
- **Startup Circuit Inactivity:** Node (1s) at the gate of switch M_5 is held at $4.195\ \mu\text{V} \approx 0\text{ V}$ by sensor M_4 . As a result, M_5 is in cutoff, verifying that the startup circuit draws zero static leakage current.

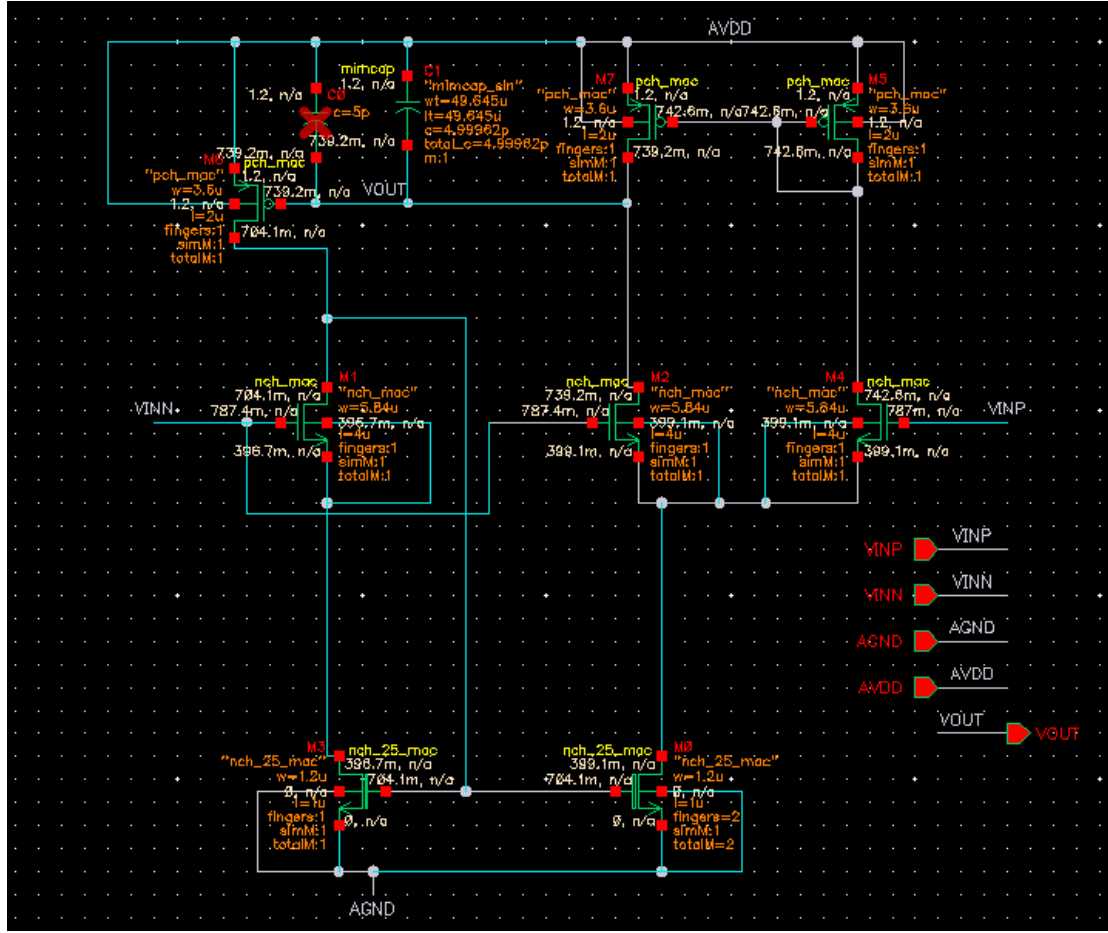


Figure 22. Detailed schematic of the self-biased OTA with the integrated mimcap, annotated with transistor operating points and node voltages.

OTA Internal Node Voltages:

As shown in Figure 22, the input common-mode voltage ($V_{INP} = 787.4 mV$, $V_{INN} = 787.4 mV$) biases the thin-oxide differential pair (M_1, M_2, M_4). The tail node settles at 399.1 mV, providing 399.1 mV of V_{DS} across NMOS current mirror devices M_0 and M_3 . This keeps the tail devices in saturation across all corners.

3. Corner Performance Across Temperature (-40°C to $+125^{\circ}\text{C}$)

The completed circuit with TSMC 65 nm physical passives was simulated across the full -40°C to $+125^{\circ}\text{C}$ temperature range under all five standard process corners (TT, SS, FF, SF, FS).

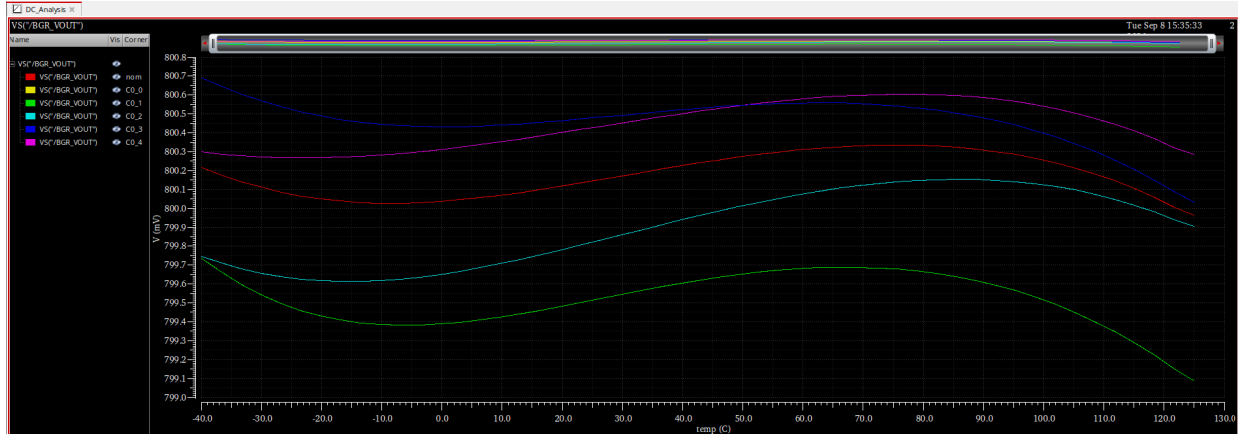


Figure 23. V_{REF} vs. Temperature (-40°C to $+125^{\circ}\text{C}$) across all process corners using TSMC 65 nm rnpoly and mimcap passives.

Corner / Condition	Peak Voltage (V_{max})	Least Voltage (V_{min})	ΔV_{REF} (Spread)	Stability Check
Nominal (TT)	800.34 mV	800.03 mV	0.31 mV	PASS (< 1 mV)
Corner 0 (SS)	800.68 mV	800.43 mV	0.25 mV	PASS (< 1 mV)
Corner 1 (FF)	799.69 mV	799.08 mV	0.61 mV	PASS (< 1 mV)
Corner 2 (SF)	800.16 mV	799.61 mV	0.55 mV	PASS (< 1 mV)
Corner 3 (FS)	800.60 mV	800.27 mV	0.33 mV	PASS (< 1 mV)

Table 9. Corner performance summary of the Bandgap Reference using physical PDK passives.

Final Performance Conclusions:

- 1. Thermal Stability with Physical Resistors:** Despite the temperature dependence of physical polysilicon, the matched ratio layout preserves zero-TC tracking. Across the entire 165°C span, every process corner exhibits a peak-to-least voltage variation strictly below **0.61 mV**, satisfying the sub-1 mV design benchmark.
- 2. Nominal Temperature Coefficient:**

$$\text{TC} = \frac{\Delta V_{REF}}{V_{nom} \cdot \Delta T} \times 10^6 = \frac{0.31 \text{ mV}}{800 \text{ mV} \times 165 \text{ K}} \times 10^6 \approx 2.35 \text{ ppm}/^{\circ}\text{C} \quad (44)$$
- 3. Global Corner Envelope:** The reference output spans from a global minimum of 799.08 mV (FF at 125°C) to a maximum of 800.68 mV (SS at -40°C). The total variation is contained within $\Delta V_{tot} = 1.60 \text{ mV}$ ($\pm 0.10\%$).
- 4. Power Specification Compliance:** Total quiescent current consumption is **7.782 μA** , meeting the $< 10 \mu\text{A}$ design requirement across all conditions.